

# Ewing sarcoma and in-utero arsenic exposure: an evidence audit

---

LEHRP · 17 September 2026 · Statement classes and source grades marked throughout. This is a research audit, not medical advice, and it establishes no causation.

## 0. The theory under audit, stated precisely

*Mothers living in Ladera Ranch during the 1999–2007 build-out were exposed to legacy arsenic (from the 1907–1912 dipping program) mobilized by grading; the exposure reached the fetus; it caused or promoted the EWSR1-FLI1 gene fusion that defines Ewing sarcoma; the initiated cells lay dormant until puberty, when the tumors emerged as the community's reported Ewing cluster.*

The chain has seven links. The audit scores each one. Summary verdict first, evidence after.

**Verdict.** The theory is *coherent* and its timing structure is *scientifically live*: a published two-step model of Ewing sarcoma proposes exactly an in-utero initiating event followed by pubertal-hormone-driven emergence, and human data prove that in-utero arsenic exposure can cause cancers that surface decades later. But two links in the chain have **no supporting literature at all** (arsenic causing the EWSR1-FLI1 fusion specifically; any Ewing excess in arsenic-exposed populations), one link has evidence pointing partly the *other* way (arsenic trioxide kills Ewing cells at drug doses), and the three Ladera-specific links (soil arsenic, maternal exposure, verified case count) are **unmeasured**. It is a testable hypothesis, not a finding, and the tests are concrete.

## 1. What Ewing sarcoma is — established science (A1/B1)

- **Definition.** A small round blue cell tumor of bone and soft tissue, defined by a chromosomal translocation fusing *EWSR1* (chr 22) to an ETS-family gene — *FLI1* in ~85–90% of cases (t(11;22)), *ERG* in most of the rest. The fusion protein EWSR1-FLI1 is an aberrant transcription factor. [A1 — pathology consensus]
- **The fusion is somatic, not inherited.** It arises in a cell of the body, not in the egg or sperm; parents do not pass Ewing sarcoma on. [A1]
- **A "quiet" genome.** Beyond the fusion, Ewing tumors carry very few mutations (recurrent secondary hits in *STAG2*, *TP53*, *CDKN2A* in subsets). This matters for the theory: there is no long mutational trail to read an environmental cause from, and the fusion itself is the event that would have to be caused. [B1]
- **Cell of origin.** Mesenchymal / neural-crest-derived progenitor cells; 2024–2025 work in zebrafish and human embryonic mesenchymal stem cells shows EWSR1-FLI1 reprograms neural-crest cells toward a mesoderm-like state and impairs DNA-damage repair. [B1 — recent, model systems]
- **How the fusion works — GGAA microsatellites.** EWSR1-FLI1 binds runs of GGAA repeats and switches on nearby genes. The length and continuity of those repeats vary between people. [B1 → A1 for the mechanism]
- **Germline susceptibility is real and large.** A GGAA-microsatellite variant near *EGR2* (10q21) and other GWAS loci raise risk; incidence is ~1.5 per million in people of European ancestry, ~0.8 in East Asian, ~0.2 in African ancestry — a 7-fold gradient explained largely by germline variation, not environment. Any Ladera analysis must adjust for the community's ancestry mix. [A1/B1]
- **Who gets it.** ~200–250 new US cases per year; peak age 10–20, median ~15; slight male excess; taller adolescents at higher risk (growth-axis signal). [A1 — SEER]
- **Presentation and diagnosis.** Persistent localized bone pain (often worse at night), swelling, sometimes fever or pathologic fracture; frequently misattributed for months to sports injury or "growing pains." Work-up: X-ray (permeative lesion, "onion-skin" periosteal reaction), MRI of the primary, biopsy with CD99 immunostaining, and **molecular confirmation of the EWSR1 rearrangement** by FISH, RT-PCR, or sequencing; staging by PET/CT and bone-marrow sampling. Treatment: multi-agent chemotherapy plus surgery and/or radiation. [A1 — clinical]

guidelines]

- **Known environmental risk factors: almost none, with one agricultural signal.** The largest case-control literature finds no confirmed environmental cause. Recurrent signals: parental farm work and pesticide handling around conception/pregnancy (Australian national study: mothers handling pesticides OR  $\approx 2$ ; pooled parental-farming analysis supportive), rural residence, hernias, and paternal wood-dust exposure. None is established as causal. [B1]

## 2. The timing claim: in-utero initiation, pubertal emergence — SUPPORTED AS A LIVE MODEL (B1)

- A 2025 developmental-origin study (Cell Reports; zebrafish + human cells) explicitly proposes a **two-step model**: an in-utero initiating event in which the fusion oncogene rewires the chromatin of a pre-neoplastic cell, followed by **postnatal acquisition of tumorigenic competence induced by high levels of pubertal hormones**. This is the theory's timing structure, in the field's own words. [B1 — model-based; not proven in patients]
- The growth axis fits: EWSR1-FLI1 depends on IGF-1 signaling; IGF-1 surges at puberty; taller adolescents carry higher risk. [B1]
- **Precedent from leukemia.** For *ETV6-RUNX1* childhood leukemia, the fusion has been proven to arise before birth by finding it on newborn blood spots and in cord blood; a 2025 study even links its newborn prevalence to a maternal pregnancy exposure (corticosteroids). That is the proof-of-concept that a fusion oncogene can be laid down in utero and surface years later. [A1/B1] **No equivalent newborn-blood-spot evidence exists yet for EWSR1-FLI1** — the timing of the Ewing fusion in real patients is unknown. [Missing evidence]
- Forward-cohort arithmetic (LEHRP, earlier): gestations during the 1999–2007 build-out reach the Ewing age peak in roughly 2012–2027. Reported Ladera dates (2013 cluster; 2023; 2024) sit inside that window. Consistent, not probative; the window is wide by construction. [Model estimate]

## 3. Arsenic: what is established, and what it would have to do here

- **Human carcinogen, IARC Group 1.** Established cancers: skin, lung, bladder; probable: liver, kidney, prostate. **Bone and soft-tissue sarcomas are not among the cancers linked to arsenic in any population.** [A1]
- **In-utero and early-life arsenic exposure causes cancer decades later — proven in humans.** Antofagasta, Chile: ~250,000 people drank high-arsenic water 1958–1970. Those exposed in utero and early childhood show elevated lung and bladder cancer and young-adult mortality; 40 years after exposure ended, odds ratios remained ~6.9 (bladder) and ~4.4 (lung). This is the single strongest support the theory has: **the general mechanism — prenatal arsenic, long latency, later cancer — is real.** [A1/B1 — Smith 2012; Steinmaus 2013]
- **Transplacental carcinogen in animals.** Brief gestational arsenic exposure in mice causes liver, lung, adrenal, ovarian and urogenital tumors in the adult offspring, and even in the next (F2) generation. [B1 — Waalkes]
- **Mechanism: not a point mutagen.** Arsenic acts through chromosomal instability (structural and numerical chromosome damage), inhibition of DNA repair, oxidative stress, and epigenetic dysregulation (DNA methylation, PRC2/histone marks). [B1]
- **Could that produce an EWSR1-FLI1 translocation?** Translocations form when double-strand breaks are mis-joined; an agent that breaks chromosomes and disables repair is *mechanistically compatible* with promoting translocations in general — and EWSR1-FLI1-expressing cells have defective DNA-damage repair, a possible cooperation. But **no published study connects arsenic to EWSR1 fusions, in cells, animals, or people.** This link is a hypothesis with a plausible mechanism and zero direct evidence. [Hypothesis]
- **Arsenic crosses the placenta** readily; fetal blood levels track maternal levels. [A1]

## 4. The counter-evidence — stated as plainly as the supporting evidence

- **No Ewing excess has been reported in the world's arsenic-exposed populations** (Chile, Bangladesh, West Bengal, Taiwan, Inner Mongolia), including cohorts exposed in utero. Caveats that weaken this as a rebuttal: Ewing is rare (~1 per million), those populations are largely non-European (baseline 3–7× lower), and no one has looked for it specifically. Absence of evidence, not evidence of absence — but a real weakness. [Missing evidence]
- **Arsenic trioxide kills Ewing sarcoma cells.** It is an FDA-approved drug (for a leukemia defined by a *different* fusion, PML-RARA), and at clinically tolerable micromolar doses it induces apoptosis in Ewing cell lines and was evaluated by the Pediatric Preclinical Testing Program with a focus on Ewing sarcoma. This cuts against a simple "arsenic feeds Ewing" story — but arsenic is famously biphasic: high doses are cytotoxic while low chronic doses promote transformation, and initiation of a fusion in a fetal progenitor is a different question from killing an established tumor. Nuanced, not fatal. [B1]
- **The dominant known risk factor is germline.** Ancestry and GGAA-repeat variants drive incidence. A cluster analysis that does not account for the community's ancestry composition is not interpretable. [A1]
- **The Ladera exposure has never been measured.** No soil, dust, water or irrigation arsenic data exist for the community; maternal arsenic body burden from 2000–2007 cannot be measured retrospectively (urine and blood clear within days; hair and nails reflect only recent months). Exposure can be *estimated* from soil data and residence history, never recovered directly. [Missing evidence]
- **The case count is unverified.** Six Ewing diagnoses in ~12 years in a community of ~25,000 would be roughly 15–20× expectation (≈1 per million per year → ~0.3 expected). An excess of that size is extraordinary, which is exactly why it must be confirmed through the California Cancer Registry before it is used for anything. Community counts can double-count, include non-residents at diagnosis, or mix tumor types. [Grade C until registry-verified]
- **The literature's own environmental hint points at agriculture broadly** — parental farming and pesticide handling — not at arsenic specifically. On former ranch land that later became an intensively landscaped suburb, that hint is compatible with legacy arsenic *and* with modern pesticides. Neutrality is required. [B1]

## 5. Link-by-link scoring

| # | Link in the chain                                                   | Status                                                                            | Grade                       |
|---|---------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------|
| 1 | A compulsory arsenical dipping program ran on this land, 1907–1912  | Verified                                                                          | A1                          |
| 2 | Legacy arsenic remains in the soil at meaningful concentrations     | Unmeasured; supported by analog sites (FL 11/12 exceedances; NSW to 14,000 mg/kg) | Hypothesis · B1 analogs     |
| 3 | Grading mobilized it into dust/water reaching residents (1999–2007) | Physically plausible (53% bare ground at peak); unmeasured                        | Inference                   |
| 4 | Pregnant residents absorbed it; the fetus was exposed               | Placental transfer established; dose unknown                                      | A1 mechanism · dose missing |
| 5 | Arsenic caused/promoted an EWSR1-FLI1 fusion in a fetal progenitor  | Mechanistically compatible; <b>no literature</b>                                  | Hypothesis                  |
| 6 | Initiated cells emerged as Ewing sarcoma at puberty                 | Published two-step model; leukemia precedent; unproven for Ewing                  | B1 model                    |
| 7 | This explains the reported Ladera cluster                           | Timing consistent; count unverified; ancestry unadjusted                          | C                           |

## 6. What would actually test it — in order of power per dollar

- **Registry confirmation (gate G01).** California Cancer Registry / SEER: verified Ewing counts, ages, and residence-at-diagnosis for the community and matched comparison communities, ancestry-adjusted. Converts grade C to A1 and answers whether there is an excess at all. Everything else depends on this.
- **Exposure measurement.** Soil arsenic (surface and depth) at the ranked targets; basin sediment core with dating; purple-pipe irrigation water; house dust in older homes. Determines whether link 2 exists.
- **Residence-timing study** through a physician/IRB pathway: for verified cases, where the mother lived during pregnancy and the child's early years, versus the grading map. Never collected by this project directly.
- **Tumor and germline genomics** (with families' consent, through their oncologists): fusion type; copy-number and chromosomal-instability profile (arsenic's expected footprint is structural damage, not a point-mutation signature); germline GGAA/EGR2 genotype to separate inherited susceptibility from environment.
- **Biomarkers of ongoing exposure** in long-term residents (toenail arsenic) — tells you about the present, not 2003, but a present signal would matter.
- **Ask the field.** The question "can environmental genotoxins initiate EWSR1-FLI1 in utero?" is open in Ewing biology; the labs that published the two-step model and the DNA-repair findings are the right people to put it to.

## 7. Approved language for conversations with physicians, attorneys and officials

- "Ewing sarcoma has no established environmental cause; its known risk is largely germline."
- "A published model proposes in-utero initiation with pubertal emergence; it is not yet proven in patients."
- "In-utero arsenic exposure is a proven cause of later cancer in humans — of the lung and bladder; no population has been examined for Ewing sarcoma."
- "Arsenic damages chromosomes and disables DNA repair, which is mechanistically compatible with producing translocations; no study links it to the Ewing fusion."
- "The reported pattern warrants investigation; the current data are insufficient to determine whether any exposure occurred or whether the case count exceeds expectation."

## Sources (held or cited)

Valery et al. 2002, *Cancer Causes & Control* (Australian national case-control; farm/pesticide exposure) · pooled parental-occupation analysis, *Int J Cancer* 2005 · Smith et al. 2012, *Environ Health Perspect* (young-adult mortality after in-utero arsenic, Antofagasta) · Steinmaus et al. 2013, *CEBP* (cancer risk 40 years after exposure cessation) · Waalkes et al. 2003–2007 (transplacental arsenic carcinogenesis, mice) · Mathieu et al. 2006, *Int J Cancer* and PPTP 2013 (arsenic trioxide vs Ewing cells) · Grünwald et al. 2015, *Nat Genet* (EGR2 GGAA microsatellite susceptibility) · Machiela et al. 2018 (Ewing GWAS) · Spector et al. 2021, *Int J Cancer* (international incidence by ancestry) · Cell Reports 2025 (neural-crest origin; two-step model) · 2025 study on newborn ETV6-RUNX1 prevalence and maternal exposure · IARC Monograph 100C (arsenic) · NCI/SEER Ewing sarcoma statistics.